

PREDICATE-TRANSFORMER- BASED VERIFICATION (LN CHAPTER 2)



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PLAN

- Hoare logic
- Verification using predicate transformers
- Verification with “Guarded Command Language” (GCL)
- Objects and exceptions
- Should at end give an answer to “how to mechanically verify a real program”.

SPECIFYING PROGRAMS

- We can use “Hoare triples” :

$$\{ \text{true} \} \text{ plusOne}(x) \text{ body } \{ \text{return} = x+1 \}$$
$$\{ \#S > 0 \} \text{ max}(S) \text{ body } \{ \text{return} \in S \wedge (\forall x: x \in S : \text{return} \geq x) \}$$

- *total* and *partial correctness* interpretation.

HOARE LOGIC

- Provides a set of “inference rules” to prove the validity of Hoare triples. Example:

$$\frac{O \Rightarrow P, \{P\} S \{Q\}}{\{O\} S \{Q\}}$$

- **Note** : LN uses the notation $\vdash O \Rightarrow P$

THE LOGIC'S GENERAL IDEA: BREAK IT DOWN!

$$\{P\} S_1 \{Q\} , \{Q\} S_2 \{R\}$$

$$\{P\} S_1 ; S_2 \{R\}$$
$$\{P \wedge g\} S_1 \{Q\} , \{P \wedge \neg g\} S_2 \{Q\}$$

$$\{P\} \text{ if } g \text{ then } S_1 \text{ else } S_2 \{Q\}$$

ASSIGNMENT

- Eventually, everything boils down to “assignments”
- An assignment is correct if

$$\frac{P \Rightarrow Q[e/x]}{\{ P \} x:=e \{ Q \}}$$

- **Note:** this assumes that the assignment only changes the value of x (it does not silently affect other variables)

LOOP

- A loop is correct if you can find an “invariant” :

$$\begin{array}{c} P \Rightarrow I \\ \{ g \wedge I \} \ S \ \{ I \} \\ I \wedge \neg g \Rightarrow Q \\ \hline \{ P \} \ \underline{\text{while}} \ g \ \underline{\text{do}} \ S \ \{ Q \} \end{array}$$

- E.g. a trivial loop:

$\{ k=1000 \} \quad \text{while } k>0 \text{ do } k := k-1 \quad \{ k=0 \}$

FEW MORE EXAMPLES

$\{ y=0 \wedge k=0 \}$ **while** $k < 10$ **do** $\{ y := y+2 ; k++ \}$ $\{ y = 20 \}$

$\{ y=0 \wedge k=10 \}$ **while** $k \neq 0$ **do** $\{ y := y+2 ; k-- \}$ $\{ y = 20 \}$

FEW MORE EXAMPLES

Home work:

$$\{ s = \text{false} \wedge k = 0 \}$$

```
while k < #b do {  
    s = s  $\vee$  b[k] ;  
    k = k + 1  
}
```

$$\{ s = (\exists i: 0 \leq i < \#b : b[i]) \}$$

PROVING TERMINATION (OF LOOP)

- Extend the previous rule to:

$$\begin{array}{l} P \Rightarrow I \\ \{ g \wedge I \} \ S \ \{ I \} \\ I \wedge \neg g \Rightarrow Q \end{array}$$
$$\begin{array}{l} \{ I \wedge g \} \ C := m ; S \ \{ m < C \} \\ I \wedge g \Rightarrow m > 0 \end{array}$$

// m decreasing

// m bounded below

$$\{ P \} \ \underline{\text{while}} \ g \ \underline{\text{do}} \ S \ \{ Q \}$$

EXAMPLE

$\{ x \geq 0 \wedge y \geq 0 \}$

```
while  $x+y > 0$  do {  
    if  $x > 0$  then {  $x--$  ;  $y := y+100$  }  
    else  $y--$   
}
```

$\{ x=0 \wedge y=0 \}$

HOME WORK

```
{ x>1 }
```

```
while x>0 do {  
    if x>1 then x := x - 2  
    else x := x + 1  
}
```

```
{ true }
```

HOARE LOGIC CANNOT BE DIRECTLY AUTOMATED

- Problem:

$$\frac{\{P\} S_1 \{Q\} , \{Q\} S_2 \{R\}}{\{P\} S_1 ; S_2 \{R\}}$$

- Let's now look at “predicate transformer-based verification”. A *predicate transformer* is a function of type :
Statement $\rightarrow$ Predicate $\rightarrow$ Predicate

FORWARD AND BACKWARD TRANSFORMER

- $\text{cp} \triangleright S P$ (forward) : transform a given pre-condition P to a post-condition.
- $\text{cp} \triangleleft S Q$ (backward) : transform a given post-condition Q to a pre-condition.
- Do they produce valid (sound) pre/post conditions? Yes if :
 $\{P\} S \{ \text{cp} \triangleright S P \}$
 $\{ \text{cp} \triangleleft S Q \} S \{ Q \}$

It follows that to prove $\{P\} S \{Q\}$, we can prove $\text{cp} \triangleright S P \Rightarrow Q$ instead. Or, we prove $P \Rightarrow \text{cp} \triangleleft S Q$ instead.

FORWARD AND BACKWARD TRANSFORMER

- Complete if:

$$\begin{array}{ll} \{P\} S \{Q\} & \text{implies} \quad \text{cp} \triangleright S P \Rightarrow Q \quad \text{valid} \\ \{P\} S \{Q\} & \text{implies} \quad P \Rightarrow \text{cp} \triangleleft S Q \quad \text{valid} \end{array}$$

- Sound and complete if :

$$\{P\} S \{Q\} \quad \equiv \quad \text{cp} \triangleright S P \Rightarrow Q$$

$$\{P\} S \{Q\} \quad \equiv \quad P \Rightarrow \text{cp} \triangleleft S Q$$

WP AND WLP TRANSFORMER

- **wp** S Q : the **weakest** pre-condition so that S terminates in Q .
- **wlp** S Q : the weakest pre-condition so that S , if it terminates, will terminate in Q .

wlp = weakest liberal pre-condition.

- Though, we will see later, that we may have to drop the completeness property of **wp/wlp**, but we will still call them **wp/wlp**.

GUARDED COMMAND LANGUAGE (GCL)

- Simple language to start
- Expressive enough to encode larger languages
- So that you can keep your logic-core simple
- Constructs
 - assignment, seq, **while**, **if-then-else**
 - **var** x **in** S , uninitialized local-var.
 - **assert** e , **assume** e
 - **try-catch**
 - program decl, program call
 - primitive types, arrays

WLP

- **wlp** skip $Q = Q$
- **wlp** $(x:=e) Q = Q[e/x]$
- **wlp** $(S_1 ; S_2) Q = \text{wlp } S_1 (\text{wlp } S_2 Q)$

We don't need to propose our own intermediate predicate!

- Example, prove:

$\{ x \neq y \} \text{ tmp} := x ; x := y ; y := \text{tmp} \{ x \neq y \}$

WLP

- **wlp** (assert e) Q = $e \wedge Q$
 - **wlp** (assume e) Q = $e \Rightarrow Q$
 - **wlp** ($S [] T$) Q = (**wlp** S Q) $\wedge$ (**wlp** T Q)
- With respect to Hoare triples these two are equivalent (any Hoare triple satisfied by one is satisfied by the other) :

if g **then** S_1 **else** S_2

$\equiv$

(**assume** g ; S_1) [] (**assume** $\neg g$; S_2)

WLP

- So, it follows:

$$\begin{aligned} \text{wlp } (\text{if } g \text{ then } S_1 \text{ else } S_2) \ Q \\ = \\ (g \Rightarrow \text{wlp } S_1 \ Q) \ \wedge \ (\neg g \Rightarrow \text{wlp } S_2 \ Q) \end{aligned}$$

- **Note** : it is equivalent to $(g \wedge \text{wlp } S_1 \ Q) \ \vee \ (\neg g \wedge \text{wlp } S_2 \ Q)$

FORMULA GROWTH

- Note that wlp of if-then-else “duplicates” Q (2x).
- So a series like:

**wlp (if g then $S1$ else $S2$;
if h then $S3$ else $S4$;
if l then $S5$ else $S5$) Q**

will duplicate Q 8x. (exponential growth in the size of the resulting wlp)

PATH-BASED VERIFICATION

- Any program S that does not contain a loop or recursion can be equivalently decomposed into linear “program paths”.
- Example :

```
if g then  $x := e_1$  else  $x := e_2$  ;  
  if h then  $y := e_3$  else  $y := e_4$ 
```

Can be decomposed to:

- assume g** ; $x := e_1$; **assume h** ; $y := e_3$
- assume g** ; $x := e_1$; **assume $\neg h$** ; $y := e_4$
- assume $\neg g$** ; $x := e_2$; **assume h** ; $y := e_3$
- assume $\neg g$** ; $x := e_2$; **assume $\neg h$** ; $y := e_4$

PATH-BASED VERIFICATION

- $\{P\} S \{Q\}$ is valid $\equiv$ forall program path σ of S : $\{P\} \sigma \{Q\}$ is valid.
- E.g. to verify :

```
      { P }  
if g then x:=e1 else x := e2 ;  
  if h then y:=e3 else y := e4  
      { Q }
```

- 
1. $\{ P \}$ **assume** g ; $x:=e_1$; **assume** h ; $y := e_3$ $\{ Q \}$
 2. $\{ P \}$ **assume** g ; $x:=e_1$; **assume** $\neg h$; $y := e_4$ $\{ Q \}$
 3. $\{ P \}$ **assume** $\neg g$; $x:=e_2$; **assume** h ; $y := e_3$ $\{ Q \}$
 4. $\{ P \}$ **assume** $\neg g$; $x:=e_2$; **assume** $\neg h$; $y := e_4$ $\{ Q \}$

Each is reducible to pred. logic formula. We can automate this!

- This approach of verification is also called “**symbolic execution**”, because it as if we symbolically execute each control path in the target program.
- The number of paths can still be a lot, but we can verify them incrementally, and even choose which ones to verify.

UNFEASIBLE PATH

- Consider as an example of program path (recall that the “assumes” came originally from branch-guards) :

$\{ P \} \text{ assume } g ; x := x + 1 ; \text{ assume } h ; y := x \quad \{ y > z \}$

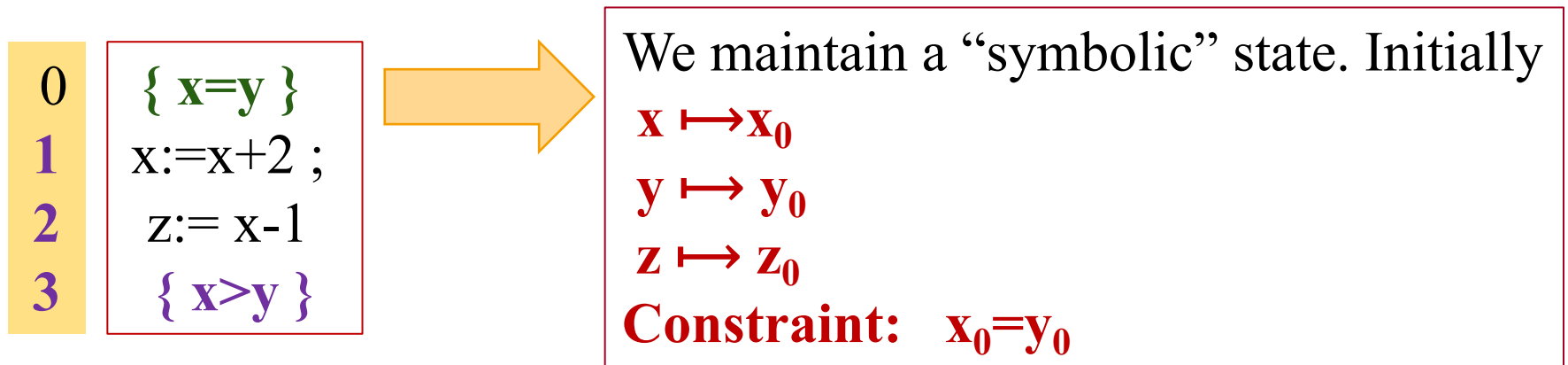
$$C_1 = P \wedge g$$

$$C_2 = P \wedge g \wedge h[x+1/x]$$

- A program path can turn out to be **unfeasible** if no actual execution can trigger it. This happens if one of its branch-guard is unfeasible towards the path pre-condition (P above) :
 - Condition g above is unfeasible if C_1 is *unsatisfiable*
 - Condition h is unsatisfiable if C_2 is *unsatisfiable*
- Verifying an unfeasible path is waste of effort, but checking if a path is unfeasible also takes effort (above, you need to check C_1 and C_2).

FORWARD SYMBOLIC EXECUTION

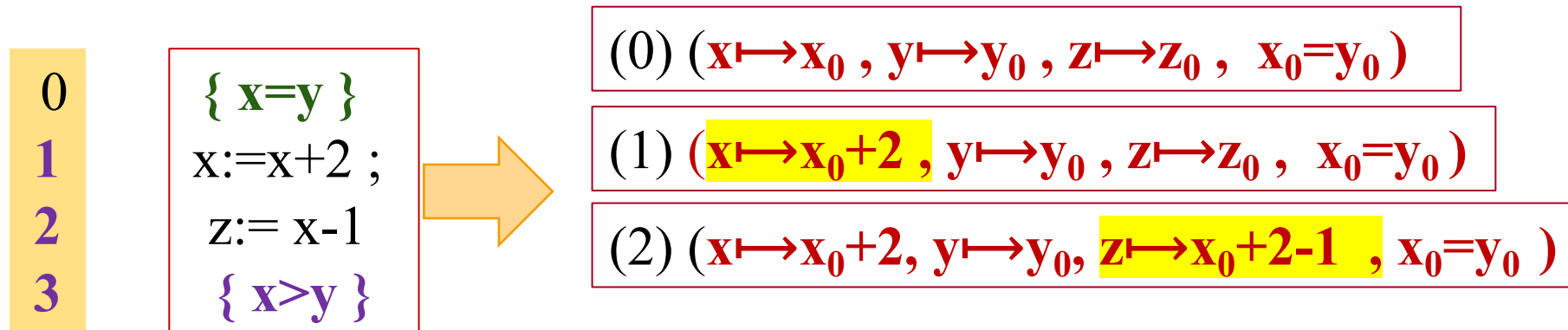
- **Forward** symbolic execution: executing a program path, using “variables” to symbolically represent all possible inputs.



- “ $x \mapsto x_0$ ” means the variable x has the value x_0 .
- x_0, y_0, z_0 : fresh variables representing initial values of x, y, z .
- “Constraint” is a condition that the symbolic values must satisfy, e.g. because it is imposed by the program’s pre-condition.

FORWARD SYMBOLIC EXECUTION

- **Forward** symbolic execution: we execute a program, taking a formula as its input :



- The Hoare triple on the left is valid if and only if the final symbolic state implies the post-condition. That is, if this is valid:

$$x=x_0+2 \wedge y=y_0 \wedge z=x_0+2-1 \wedge x_0=y_0 \implies x>y$$

FORWARD SYMBOLIC EXECUTION

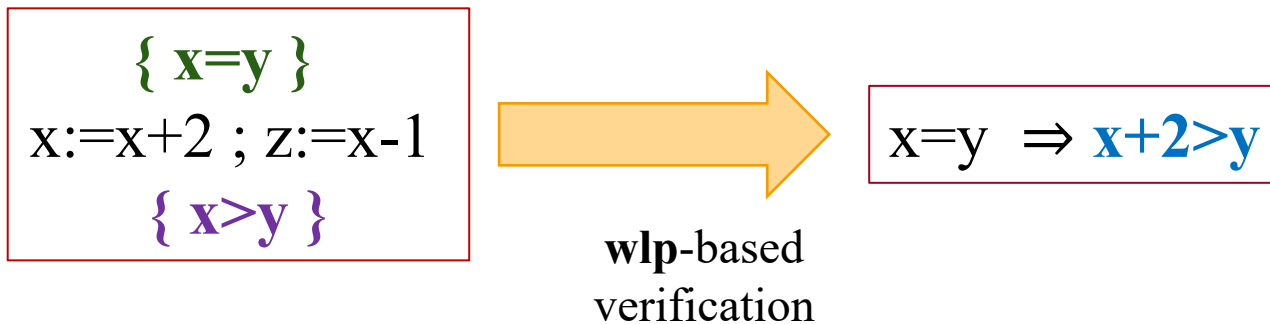
- More generally a symbolic state consist of a pair (s, C) where C is a constrain and s is a mapping $x \mapsto e$ that maps every program variable to its symbolic value (which can be an expression e.g. $\alpha+1$). To keep it simple, we won't keep track of a stack (needed if you have recursive calls).
- Initially the symbolic state is $([x_1 \mapsto \alpha_1, x_2 \mapsto \alpha_2, \dots], true)$ where α_k is a fresh variable representing an arbitrary initial value of x_k .
- Convert the starting pre-condition P to an stmt **assume** P .
- Given a program path ρ (sequence of instructions), and an initial symbolic state (s, C) , to symbolically execute ρ we simply execute the instructions in ρ in the order as they appear, passing on the new symbolic state after an instruction ρ_k to the execution of the instruction ρ_{k+1} .

FORWARD SYMBOLIC EXECUTION

- executing **assume** P on (s, C) gives $(s, P \wedge C')$ as the new symbolic state. C' is obtained by replacing every program variable z that occurs in C with $s(z)$ (the symbolic value of z in the state s).
- executing **assert** Q on (s, C) gives the same state (s, C) as the new symbolic state, if $C \wedge s \Rightarrow Q$ holds. Else the execution is aborted, reporting an error (the asserted Q is violated).
- executing **$y := e$** on (s, C) gives (**update** y e' s , C) as the new symbolic state, where e' is the symbolic value of the expression e on the state s , which can be obtained by replacing every program variable z that occurs in e with $s(z)$ (the symbolic value of z in the state s).

BACKWARD VS FORWARD SYMBOLIC EXECUTION

- Consider again the verification of:



- Wlp-based. The program is valid iff $x=y \Rightarrow x+2>y$
- Forward symbolic execution: the program is valid iff

$$x=x_0+2 \wedge y=y_0 \wedge z=x_0+2-1 \wedge x_0=y_0 \Rightarrow x>y$$

BACKWARD VS FORWARD TRANSFORMATION

- Backward execution yields: $x=y \Rightarrow x+2>y$.
- Forward execution yields:

$$y=y_0 \wedge x_0=y_0 \wedge x = x_0+2 \wedge z=x_0+2-1 \Rightarrow x>y$$

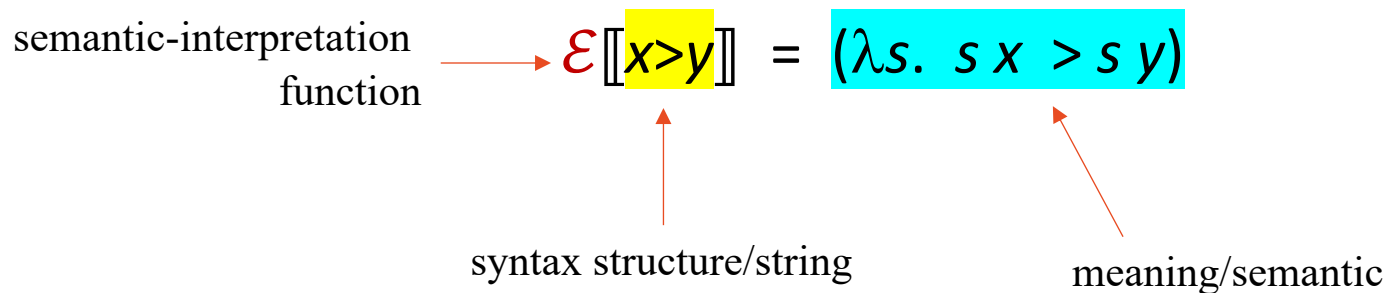
- Backward transformation: yields cleaner formulas, containing only conditions relevant towards the post-cond.
- Forward transformation
 - The direction is more intuitive
 - The intermediate formulas also produce conditions that correspond to the feasibility of each branch-guard (1x, the blue box above), so you can immediately check the guard feasibility. Note we can also check this via wlp; it is just that we need more staging in the corresponding symbolic execution.

CAN WE PROVE THE SOUNDNESS AND COMPLETENESS OF THE WLP TRANSFORMER ?

- Yes, e.g. wrt a denotational semantic. I will just give a sketch of how to do this.
- Consider the following domains as a simple model of program states:
 - **val** : the space of all possible values of program variables
 - A state is modelled as a **function** from variable-names to **val**. So, if s is a state, “ $s\ x$ ” is the value of variable x in the state s .
 - **State** : the space of all possible program states.
- Note: Chapter 1 and 2 propose two different representations of states. They are both usable, as they both support state query and update.

THE SEMANTIC OF EXPRESSIONS AND PREDICATES

- $\mathcal{E} : \text{expr} \rightarrow (\text{State} \rightarrow \text{val})$
- $\mathcal{E} : \text{Pred} \rightarrow (\text{State} \rightarrow \text{bool})$ // overloading $\mathcal{E}$
- e.g.



THE SEMANTIC OF EXPRESSIONS AND PREDICATES

- Note that $P : \mathbf{State} \rightarrow \mathbf{bool}$ can equivalently be seen as a set of all the states on which P is true:

$$“P \text{ as a set}” = \{ s \mid s \in \mathbf{State} \wedge P s \}$$

- Predicate operators translate to set operators, e.g. $\wedge, \vee$ to $\cap, \cup$, negation to complement wrt $\mathbf{State}$.
- This ordering: “ $P \Rightarrow Q$ is valid” translates to $P \subseteq Q$

THE SEMANTIC OF STATEMENTS

- We model a statement as a function of type **State** $\rightarrow$ **State**.

$$\mathcal{S} : \text{stmt} \rightarrow (\text{State} \rightarrow \text{State})$$

- Alternatively: $\mathcal{S} : \text{stmt} \rightarrow (\text{State} \rightarrow \text{Pow}(\text{State}))$ to allow non-determinism.

where **Pow(State)** is the domain of all subsets of **State**. On this domain, we have: $\cap \cup \subseteq \supseteq$

THE SEMANTIC OF STATEMENTS (DETERMINISTIC)

- $\mathcal{S} \llbracket \text{skip} \rrbracket = (\lambda s. s)$
- $\mathcal{S} \llbracket x := e \rrbracket = (\lambda s. \text{update } s \ x \ (\mathcal{E} \llbracket e \rrbracket s))$
- $\mathcal{S} \llbracket S_1 ; S_2 \rrbracket = (\lambda s. \mathcal{S} \llbracket S_2 \rrbracket (\mathcal{S} \llbracket S_1 \rrbracket s))$
- We can now “execute” a program over our model-states.
We can also study properties about programs by studying their semantical models.

SEMANTIC OF HOARE TRIPLE (DETERMINISTIC)

- $\mathcal{H} : \text{Hoare-triple} \rightarrow \text{bool}$

$$\mathcal{H}(\{P\} S \{Q\}) = \forall s: \mathcal{E}[\![P]\!] s : \mathcal{E}[\![Q]\!] (\mathcal{S}[\![S]\!] s)$$

- wlp is sound and complete if:

$\{P\} S \{Q\}$ if and only if $P \Rightarrow \mathbf{wlp} S Q$ is valid

- It comes down to proving that: for all state s , and post-cond Q :

$$\mathcal{E}[\![Q]\!] (\mathcal{S}[\![S]\!] s) \text{ if and only if } \mathcal{E}[\![\mathbf{wlp} S Q]\!] s$$

Can be proven inductively over the structure of S .

BACK TO DEALING WITH LOOP

- Unfortunately, no general way to calculate **wlp** of loops.
- For annotated loop, let's “define” :

wlp (inv I while g do S) Q = I , provided

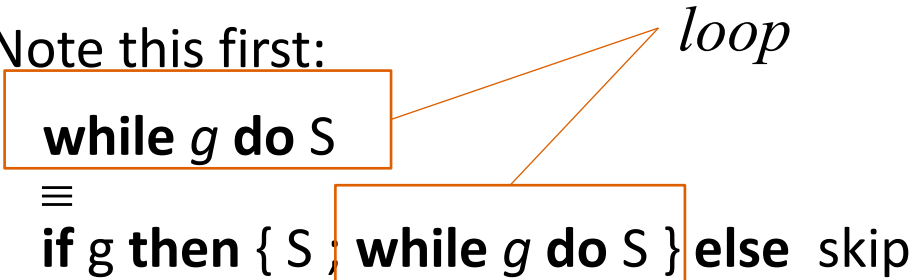
- $I \wedge \neg g \Rightarrow Q$
- $I \wedge g \Rightarrow \text{wlp } S I$

Note that this gives a sound pre-cond, but does not necessarily give the weakest pre-cond.

WHAT IF THERE IS NO I ANNOTATED ?

- Heuristics to construct invariants e.g. based on the form of the post-condition → not in scope.
- Dynamically infer invariants → not in scope.
- Ask an LLM → not in scope.
- Non-heuristic approaches, simple; can be used as starting points.
 - Fix point based
 - Unfolding

WLP AS FIX POINT

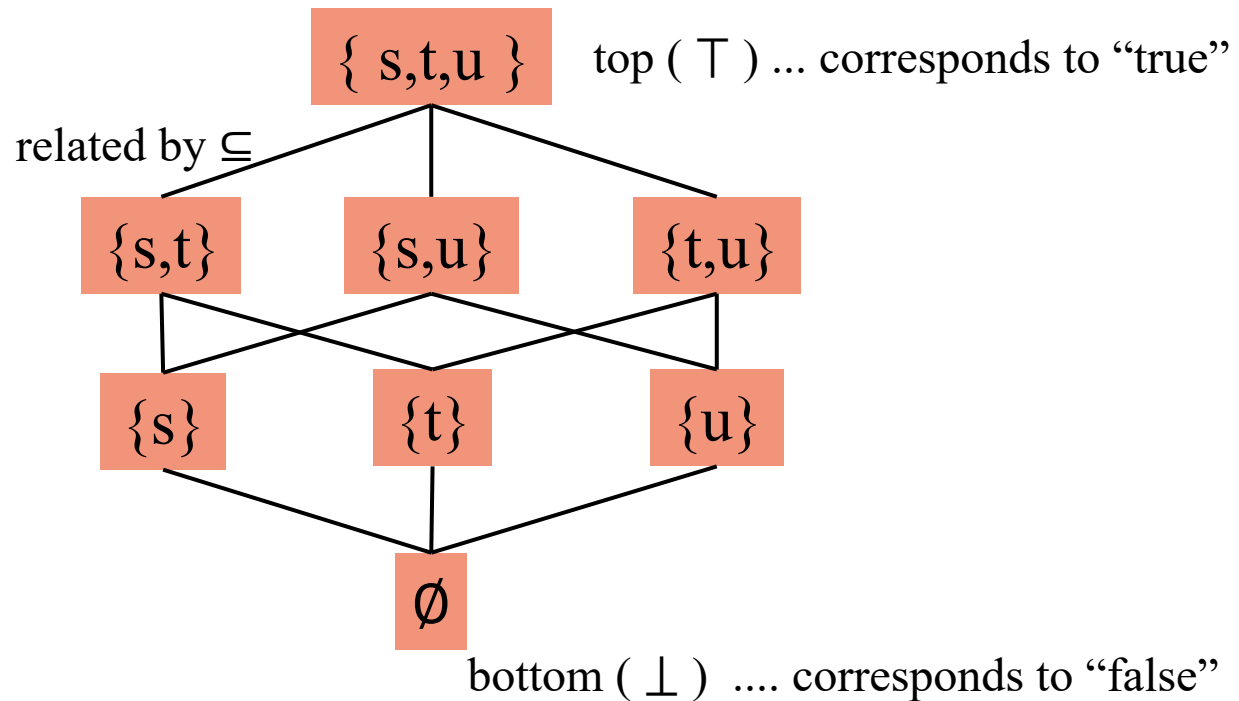
- Note this first:
 $\text{while } g \text{ do } S$
 $\equiv$
 $\text{if } g \text{ then } \{ S ; \text{while } g \text{ do } S \} \text{ else skip}$
- 

- let $W = \text{wlp loop } Q$.
 $= (g \Rightarrow \text{wlp } S (\text{wlp loop } Q)) \wedge (\neg g \Rightarrow Q)$
 $= (g \wedge \text{wlp } S (\text{wlp loop } Q)) \vee (\neg g \wedge Q)$
 $\quad \quad \quad \underbrace{\hspace{10em}}_W$
 $\quad \quad \quad \underbrace{\hspace{15em}}_{\mathbf{F}(W)}$

- We are looking for the “weakest” solution of $W = \mathbf{F}(W)$

THE DOMAIN OF STATE PREDICATES, $\text{Pow}(\Sigma)$

- Suppose Σ (set of all states) = $\{s, t, u\}$. The domain $\text{Pow}(\Sigma)$, ordered by $\subseteq$:



$X \cup Y$ (union) acts as the least upper bound of X and Y

$X \cap Y$ (intersection) acts as the greatest lower bound of X and Y

SOME BIT OF FIX POINT THEORY

- $(A, \leq)$ where $\leq$ is a p.o., is a *complete lattice*, if every subset $X \subseteq A$ has a supremum and infimum.
 - Supremum = least upper bound = join $\bigvee X \quad \bigsqcup X$
 - Infimum = greatest lower bound = meet $\bigwedge X \quad \bigsqcap X$
 - So, we will also have the supremum and infimum of the whole A , often called top $\top$ and bottom $\perp$.
- Let $f: A \rightarrow A$. An x such that $x = f(x)$ is a *fix point* of f .
- **Knaster-Tarski.** Let $(A, \leq)$ be a complete lattice. If $f: A \rightarrow A$ is monotonic over $\leq$, and P is the set of all its fix points, then P is non-empty, and $(P, \leq)$ is a complete lattice.
- (thus, both $\bigsqcup P$ and $\bigsqcap P$ are also in P).

SOME BIT OF FIX POINT THEORY

- The domain $(\mathbf{pow}(\Sigma), \subseteq)$ is a complete lattice.
 - Σ = top
 - $\emptyset$ = bottom
 - $A \cup B$: least upper bound ($\bigvee$)
 - $A \cap B$: greatest lower bound ($\bigwedge$)
- So, if $\mathbf{F}$ defined before is monotonic within this domain, then it has a greatest fix-point.
- Is $\mathbf{F}$ monotonic ?
- Is $\mathbf{wlp}$ monotonic ?

SOME BIT OF FIX POINT THEORY

- Consider now $f : \mathbf{pow}(\Sigma) \rightarrow \mathbf{pow}(\Sigma)$. It is $\cap$ -continuous if for all decreasing chain $X_0 \supseteq X_1 \supseteq X_2 \dots$:

$$f(X_0 \cap X_1 \cap \dots) = f(X_0) \cap f(X_1) \cap \dots$$

- If f is $\cap$ -continuous, it is also monotonic.
- Is **wlp** $\cap$ -continuous ?

FP ITERATION

- *Define:*
 - $f^0(X) = X$,
 - $f^{k+1}(X) = f(f^k(X))$
- Suppose f is $\cap$ -continuous. Consider the series
 $f^0(\Sigma), f^1(\Sigma), f^2(\Sigma) \dots$

Corollaries:

- f is also monotonic.
- The series is a decreasing chain.
- $\alpha = f^0(\Sigma) \cap f^1(\Sigma) \cap f^2(\Sigma) \cap \dots$ is a fix point of f .
- α is the greatest fix point of f .

FP ITERATION

- How to compute $\cap \{ f^k(\Sigma) \mid k \geq 0 \}$?
 - compute $f^0, f^1, f^2, \dots$ but notice you only need to keep track of the last.
 - $X := \Sigma$;
 while $X \neq f(X)$ **do** $X := f(X)$
- Will give the greatest FP, if it terminates.
- For **wlp** :
 - $W := \text{true}$;
 while $W \neq \mathbf{F}(W)$ **do** $W := \mathbf{F}(W)$

where $\mathbf{F}(W) = (g \wedge \text{wlp } S \ W) \vee (\neg g \wedge Q)$

EXAMPLE 1

while $y > 0$ **do** $\{ y := y - 1 \} \quad \{ y = 0 \}$

- Q is $y = 0$
- $W_0 = \text{true}$
- $$\begin{aligned} W_1 &= (y > 0 \wedge \mathbf{wlp} \ S \ W_0) \vee (\neg(y > 0) \wedge y = 0) \\ &= (y > 0 \wedge \text{true}) \vee (y = 0) \\ &= y \geq 0 \end{aligned}$$
- $$\begin{aligned} W_2 &= (y > 0 \wedge y - 1 \geq 0) \vee (\neg(y > 0) \wedge y = 0) \\ &= y \geq 1 \vee y = 0 \\ &= y \geq 0 \end{aligned}$$
- $W_2 = W_1$

EXAMPLE 2

while $y > 0$ **do** { $y := y - 1$ } { d }

- Q is d -- c,d are bool vars
- $W_0 = \text{true}$
- $W_1 = (y > 0 \wedge \mathbf{wlp} \ S \ W_0) \vee (\neg(y > 0) \wedge d)$
 $= (y > 0 \wedge \text{true}) \vee (y \leq 0 \wedge d)$
- $W_2 = (y > 0 \wedge \mathbf{wlp} \ S \ W_1) \vee (y \leq 0 \wedge d)$
 $= (y > 0 \wedge y > 1) \vee (y = 1 \wedge d) \vee (y \leq 0 \wedge d)$
- $W_3 = (y > 2) \vee (y = 2 \wedge d) \vee (y = 1 \wedge d) \vee (y \leq 0 \wedge d)$
- does not terminate ☹

EXAMPLE 2

```
while  $y > 0$  do {  $y := y - 1$  } {  $d$  }
```

-
- $W_3 = (y > 2) \vee (y = 2 \wedge d) \vee (y = 1 \wedge d) \vee (y \leq 0 \wedge d)$
- does not terminate ☹️
- **But** we know that $W_3 \supseteq$ the actual fix-point (the wlp).
- We can take W_3 as the "wlp" of the loop. If P is the given pre-cond of the loop:
 - $P \Rightarrow W_3$ turns out to be not valid $\rightarrow$ we find a bug!
 - $P \Rightarrow W_3$ turns out to be valid $\rightarrow$ we still don't know if the loop is correct

EXAMPLE 3

while $y > 0$ **do** { **assert** d ; $y := y - 1$ } { d }

- Q is d -- c, d are bool vars
- $W_0 = \text{true}$
- $$\begin{aligned} W_1 &= (y > 0 \wedge \mathbf{wlp} \ S \ W_0) \vee (\neg(y > 0) \wedge d) \\ &= (y > 0 \wedge d) \vee (y \leq 0 \wedge d) \\ &= d \end{aligned}$$
- $$\begin{aligned} W_2 &= (y > 0 \wedge \mathbf{wlp} \ S \ W_1) \vee (y \leq 0 \wedge d) \\ &= (y > 0 \wedge d) \vee (y \leq 0 \wedge d) \\ &= d \end{aligned}$$
- $W_2 = W_1$

FINITE UNFOLDING

- Define

$$[\mathbf{while}]^0 (g, S) = \mathbf{assert} \neg g$$
$$[\mathbf{while}]^{k+1} (g, S) = \mathbf{if} \ g \ \mathbf{then} \ \{ S ; [\mathbf{while}]^k (g, S) \} \\ \mathbf{else} \ \mathbf{skip}$$
$$\langle \mathbf{while} \rangle^0 (g, S) = \mathbf{assume} \neg g$$
$$\langle \mathbf{while} \rangle^{k+1} (g, S) = \mathbf{if} \ g \ \mathbf{then} \ \{ S ; \langle \mathbf{while} \rangle^k (g, S) \} \\ \mathbf{else} \ \mathbf{skip}$$

- $[\mathbf{while}]^k$ iterates at most k -times, then stops.
- $\langle \mathbf{while} \rangle^k$ iterates at most k times; higher iterations are ignored/removed.

REPLACING WHILE WITH [WHILE]^K

$\{ P \} \text{ while } y > 0 \text{ do } \{ y := y - 1 \} \{ y = 0 \}$

■ **wlp** ($[\text{while}]^2 (y > 0, y := y - 1)$) ($y = 0$)

$$= (y = 2 \vee y = 1 \vee y = 0)$$

- Works if P says y is exactly that $(0, 1, 2)$.
- Does not work if P is e.g. $y = 3$ or $y \geq 0$
- Such unfolding produces *sound* wlp, but *incomplete*.

REPLACING WHILE WITH $\langle \text{while} \rangle^K$

- **wlp** $(\langle \text{while} \rangle^2 (y > 0, y := y - 1)) \quad (y = 0 \wedge b)$

$$= y > 2 \vee (y = 2 \wedge b) \vee (y = 1 \wedge b) \vee (y = 0 \wedge b)$$

- $P : y = 0 \vee y = 1 \dots$ works
- $P : y \geq 0 \dots$ “works” as well
- This unfolding yields wlp which is complete but unsound. So, if $P \Rightarrow W$, W is the above wlp, is valid, we don't know if the original specification is also valid. However, if $P \Rightarrow W$ is invalid, then so is the orig. spec.

VERIFICATION OF OO PROGRAMS

- GCL is not OO, but we can encode OO constructs in it. We'll need few additional ingredients :
 - local variables
 - simultant assignment
 - program call
 - array
 - object
 - method

LOCAL VARIABLE

- **wlp** (**var** x **in** $x:=1$; $y := y+x$ **end**) ($x=y$)

Rename loc-vars to fresh-vars, to avoid captures:

wlp (**var** x' **in** $x':=1$; $y := y+x'$ **end**) ($x=y$)

- Let's try another example :

wlp (**var** x' **in** **assume** $x'>0$; $y := y+x'$ **end**) ($x=y$)

- Note that if x' is fresh we can alternatively treat this as:

assume $x'>0$; $y := y+x'$

SIMULTANT ASSIGNMENT

- For example: $x, y := y, x+y \quad \{x=y\}$

$$\text{wlp} (x, y := e_1, e_2) Q = Q[e_1, e_2 / x, y]$$

- Compare it with : $(x=y) [y/x] [x+y/y]$
- But e.g. this is **not allowed**: $x, x := e_1, e_2$

PROGRAM AND PROGRAM CALL

- Syntax: $Pr(x, y \mid r) \text{ body}$
- x, y are input parameters $\rightarrow$ passed by value
- r is an output parameter.
- Syntax of program call: $a := Pr(e_1, e_2)$

Example :

- $inc(x \mid r) \{ r := x+1 \}$
- $x := inc(x)$
- $y := inc(x)$

ENCODING PROGRAM CALL

- Consider : $Pr(x, y \mid r) \text{ body}$
- We treat program call as syntactic sugar :

$$a := Pr(e_1, e_2)$$
$$\equiv$$
$$\mathbf{var} \ x, y, r \ \mathbf{in} \ x, y := e_1, e_2 ; \text{ body} ; a := r \ \mathbf{end}$$

Rename to make the loc-vars fresh.

- This allows you to using existing rules to calculate the wlp of calls (except when we have recursion, see next slide).
- example:

$$\mathbf{wlp} \ (x := \text{inc}(x)) \ (x > 1) \rightarrow x + 1 > 1$$

BLACK-BOX CALL

- What if we don't know the body? Assuming you still have the specification, e.g. :

$\{ x > 0 \} \text{ drop}(x \mid r) \{ r < x \}$

We treat this as having this body :

$\text{drop}(x \mid r) \{ \text{assert } x > 0 ; \text{assume } r < x \}$

- Note:** reference to input params (e.g. “x”) in a method post-cond is intended to refer to their initial value.
- example: **wlp** $(x := \text{drop}(x)) \ (x < 7)$
 $\rightarrow x > 0 \wedge (r < x \Rightarrow r < 7)$
- (assuming r is fresh!)
- Note:** this allows us to handle call to recursive programs, if they provide specifications (but proving the correctness of a recursive program is still another problem to solve)

ARRAYS

- Arrays in this GCL are infinite (but not in your project).
- $a := b$ is assumed to clone b into a .
- Introduce the notation $a(i \text{ **repby** } e)$ to denote a clone of the array a , but differs at i -th element, which now has the value e . See LN, Rule 2.6.12.
- Treat $a[i] := e$ as $a := a(i \text{ **repby** } e)$
- Example :
 wlp $(a[i] := 0) (a[i] = a[3])$
- Encode $a[i] := 0$ in an RL language e.g. as :
 assert $0 \leq i < \#a ; a[i] := 0$

ARRAY ASSIGNMENTS TRIGGER CASES

- Consider wlp calculation of:

$$(k+1=1 \rightarrow 2 \mid (k+1=k \rightarrow 1 \mid a[k+1])) = 2$$

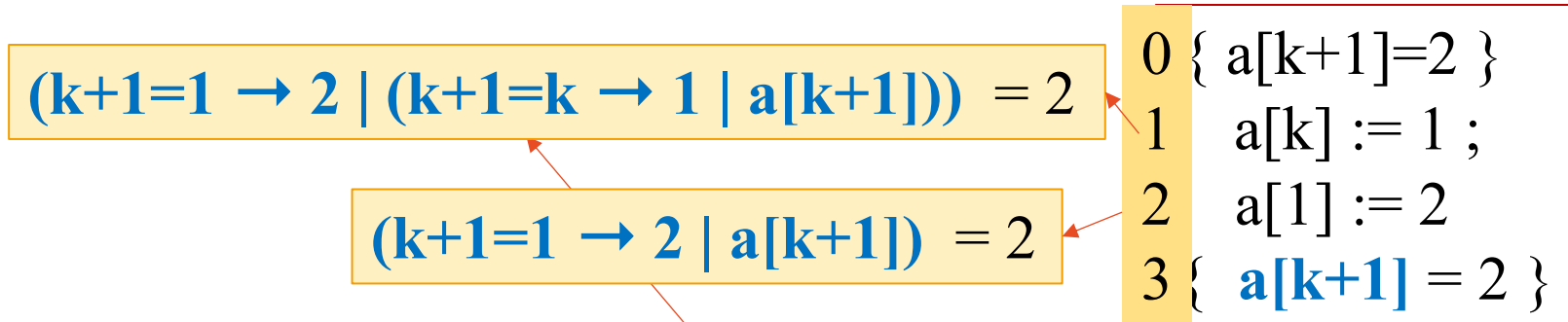
$$(k+1=1 \rightarrow 2 \mid a[k+1]) = 2$$

0	{a[k+1]=2}
1	a[k] := 1 ;
2	a[1] := 2
3	{ a[k+1] = 2 }

- For each array expression in the post-condition, each array assignment adds a nested conditional expression. Above this leads to 3 cases that the back-end prover must consider. If the post-cond has one more array expression, it will create its own 3 cases. In combination with the first, now there are 3x3 cases to consider!

ARRAY ASSIGNMENTS TRIGGER CASES

- Consider again this wlp calculation of:



- We can reduce/simplify **this**
- Simplifying cost effort.

OBJECTS

- Introduce 'heap' $H : [\text{int}][\text{fieldname}] \rightarrow \text{value}$
- We may want to introduce one heap for each Class; but let's ignore this here.
- N , representing the number of existing objects.
- Objects are stored in $H[0] \dots H[N-1]$
- Two types of values : primitive values, or reference to an object.
- Encoding :
 - $x := o$, unchanged, but note o is thus an int
 - $o.\text{fn} := 3$ is encoded by $H[o][\text{fn}] := 3$
 - Suppose C has a single int-field named a
 $x = \text{new } C()$ is encoded by $\{ H[N][a] := 0 ; x := N ; N++ \}$

METHODS

- Let $m(x|r)$ be a method of class C. It implicitly has the instance-object and the whole heap as input and output parameter.
- Example: **class** C { a:int
 inc(d) { **this**.a := **this**.a+d ; } }

The method is translated to:

$$\text{inc}(\text{this}, H, d \mid H') \{ H[\text{this}][a] := H[\text{this}][a] + d ; \\ H' := H \}$$

- call o.inc(3) is translated to ... ?

EXCEPTION

- Exception introduces non-standard flow of execution, which are often error prone.
- The problem is, exception can be thrown from many points in the program, basically exploding the goals to solve for verification.
- But first ... how to deal with exception in Hoare logic / **wlp** ?
 - Approach-1 : introduce a variable *exc* to represent that the program has entered an exceptional state, and explicitly encode the flow in the **wlp**.
 - Approach-2: extend post-condition to be a pair (Q_n, Q_e) representing the desired situation when a program terminates normally, and when it terminates exceptionally.
 - Approach-3: represent a program as a control flow graph

APPROACH 1

- Introduce a global variable *exc* : bool, initially false
- **raise** sets this variable to true
- entering a handler resets it to false
- A state where *exc* is true, is an ‘exceptional’ state, else it is a normal state.
- Multiple exception types can be encoded, but let’s ignore this here...
- The obvious:
 wlp raise $Q = Q[\text{true}/exc]$

APPROACH 1

- $S_1 ; S_2$ is treated as: $S_1 ; \text{if exc then skip else } S_2$

Downside: this blows up the resulting wlp since each “;” will add one conditional clause.

We can optimize this by statically checking if exc would be true after S_1 . If so, there is no need to expand S_2 . Examples:

- S_1 does not contain any raise stmt.
- S_1 is enclosed by an exception handler, that does not contain raise.
- **while** also need to be transformed due to the implicit sequencing between iterations

ASSIGNMENTS AND GUARDS

- In GCL evaluating an expression does not crash; so you either have to **insist** that they don't. E.g. :

$x := a[k]/y$

is transformed to:

```
assert  $0 \leq k < \#a$  ;  
assert  $y \neq 0$  ;  
 $x := a[k]/y$  ;
```

- Similar situation with guards in ite and loop.

ASSIGNMENTS AND GUARDS

- Or we model exception throwing executions. E.g. :

$x := a[k]/y$

is then transformed to :

```
if  $k < 0 \vee k \geq \#a$  then raise ;  
if  $y = 0$  then raise ;  
 $x := a[k]/y$  ;
```

- Similar situation with guards in ite and loop.

EXCEPTION HANDLER

- **S ! H (try S catch H)**

is treated as:

S ; if exc then { exc:=false ; H } else skip

(";" should not be expanded here)

APPROACH 2

- Extend the post-condition from a single predicate to a pair of predicate:

$\{P\} S \{Q,R\}$ where Q describes the expectation when S terminates normally, and R the expectation when S terminates by exception.

- **wlp** $(x := e) (Q,R) = Q[e/x]$ -- assuming e does not throw an exception.
- **wlp** **raise** $(Q,R) = R$
- **wlp** $(S_1 ; S_2) (Q,R) = \text{wlp } S_1 (\text{wlp } S_2 (Q,R), R)$
- **wlp** $(S ! H) (Q,R) = \text{wlp } S (Q, \text{wlp } H (Q,R))$
- This avoids blow up that we saw in Approach-1, but on the other hand wlp becomes more complicated.

EXCEPTION

- Approach 1:
 - wlp-rules are still the same
 - But it leads to blow up, unless we do the optimization.
- Approach 2:
 - does not blow up, but we need to keep track of two post-conditions.
- To consider: using a control flow graph.
 - But wlp is then no longer syntax driven.

SUMMARY

- You have now the full **wlp**-based logic to verify GCL programs.
- The calculation of **wlp** is syntax driven.
- If loops are annotated, the calculation of **wlp** is fully automatic, else you may have to do some trade off.
- RL languages can be translated to GCL; we have shown how some core OO constructs can be translated.
- **wlp**-based logic does *not* deal with concurrency.